06. Model Selection

Explain why the pipeline uses different model families for different tasks.

Project: Automated Mobile-Technology Brand Sentiment and Headline Generator

Source archive: NLP-Sentiment-Classification-of-Mobile-brands-main.zip | Notebook: main.ipynb | Supporting report and presentation included

Theme classification: TF-IDF + XGBoost

TF-IDF weights a term by its frequency in a document and rarity across the corpus. It reduces the influence of ubiquitous terms and emphasizes discriminative words. XGBoost builds an additive ensemble of decision trees, fitting successive trees to correct previous errors. The team selected this lighter model because theme filtering processes roughly four times as much data as downstream tasks.

Brand identification: regex dictionary

The known brand universe is relatively small (about 150 names), so deterministic pattern matching is fast, transparent and easy to update. NER would be more flexible for open-world entities but requires annotated data and can confuse brand names with ordinary words or product terms.

Sentiment: contextual models and efficient fallbacks

The report explores BERT sequence-pair classification using the brand name and text, contextual embeddings around the brand, dependency relations between brands and adjectives, and a lexicon score weighted by token distance to the brand. It also uses TF-IDF classifiers for single-brand cases to reduce latency and an ensemble to produce final sentiment. The notebook loads a multilingual BERT sentiment checkpoint and calls a SentimentClassifier; exact model internals live in omitted modules.

Headline generation: T5-base

T5 is an encoder-decoder text-to-text Transformer fine-tuned to generate an English headline from an English-translated article. The report says it was fine-tuned for two epochs, with early stopping based on validation ROUGE and BLEU. PEGASUS and a Lead-3 extractive baseline were tested; T5-base reportedly performed better. mT5 on noisy untranslated multilingual text performed worse than T5 on translated English data.

Model-selection trade-offs

Task	Chosen approach	Reason / trade-off
Theme	TF-IDF + XGBoost	Fast, compact, adequate for binary filtering
Brand	Regex over curated dictionary	Small known vocabulary; deterministic and fast
Sentiment	BERT/context + TF-IDF fallback + ensemble	Context-sensitive, target-specific; higher latency
Headline	Fine-tuned T5-base	Abstractive generation; English translation required

Interview note

> The central engineering insight is a cascade: use cheaper models to filter the full stream, then spend more compute only on mobile-tech content and difficult multi-brand sentiment cases.